

DECISION LOG

Team: Bug Busters (HM26-1E71) **Product:** AURA Learn **Sub-problem:** Adaptive Learning & Real-Time Intervention **Date:** 27 September 2026

Q1. What approach did we take, and what did we reject?

Our approach: AURA Learn uses a deterministic adaptive-learning core built around a prerequisite knowledge graph, mastery tracking, adaptive difficulty, struggle detection, interest-based AI contextual re-theming, virtual labs, and human-in-the-loop facilitator intervention. Topics are connected through prerequisite relationships, so a learner's mastery of foundational concepts influences access to dependent concepts. The mastery model combines quiz accuracy, recent performance, prerequisite mastery, retention, and virtual-lab performance. Struggle detection considers repeated errors, low accuracy, excessive time, hint dependency, topic revisits, skipped questions, and prerequisite weakness. Runtime AI is restricted to the narrative/context layer: generated educational content must preserve the numerical values, formulas, answers, learning objective, concept, and difficulty. The facilitator remains the final decision-maker for intervention.

Alternative considered and rejected: an end-to-end LLM tutor that would autonomously judge mastery, select the next curriculum concept, determine progression, and replace the facilitator in the intervention loop. We rejected this because core learning state and progression need to remain transparent, predictable, auditable, and academically safe.

Q2. Why did we reject it? The trade-off

Dimension	Our approach — AURA Learn	Rejected — Full LLM Tutor
Learning consistency	Deterministic prerequisite gating and transparent mastery rules make progression explainable and reproducible.	LLM-based mastery and progression can be variable and harder for educators to audit.
Academic integrity	Structured academic content remains authoritative; AI changes context only, with validation and fallback.	A generative model can introduce incorrect values, formulas, answers, or reasoning.
Teacher role	The system detects and recommends; the facilitator reviews and decides the intervention.	An autonomous tutor could reduce teacher visibility into why and how intervention decisions are made.
Reliability	Core progression, mastery, struggle and intervention behavior do not need to depend on an LLM call.	A network, API, or model failure could directly affect core learning decisions.

Trade-off we accepted: this architecture requires explicit curriculum and prerequisite modeling and deterministic learning rules up front. We accepted that engineering cost because the core value of AURA depends on explainable progression, measurable struggle, academic correctness, and meaningful facilitator intervention.

Q3. What breaks at scale, and how would we address it?

What becomes difficult first	Why	How we'd address it
Student-state write load	As the number of learners grows, frequent attempts, mastery updates and intervention-state changes create increasing database writes.	Separate high-frequency learning telemetry from durable mastery state, introduce asynchronous processing, and batch persistence while retaining PostgreSQL/Supabase as the durable source of truth.
Offline / intermittent connectivity	Learners may complete activities with weak connectivity, while runtime AI re-theming requires an external model connection.	Cache core learning content locally, queue learner events, reconcile them deterministically after reconnect, and fall back to original content when the AI service is unavailable.
Facilitator alert overload	A large class can generate many simultaneous struggle signals, making individual alerts difficult for one facilitator to act on.	Aggregate related signals into topic/cohort summaries and prioritize high-risk students with clear, actionable recommendations.

First scale change: separate high-frequency learning telemetry from durable mastery state and introduce asynchronous processing. This preserves the current Supabase/PostgreSQL architecture for the prototype while providing a clear path toward larger deployments.

Self-check: ✓ One approach and one rejected alternative ✓ Trade-off admitted ✓ Concrete scaling risks and remedies included ✓ Claims aligned with the AURA Learn PRD and submission README.